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Biomass-based Acidic Deep Eutectic Solvents for Efficient Dissolution of Lignin: Towards Performance and Mechanism Elucidation

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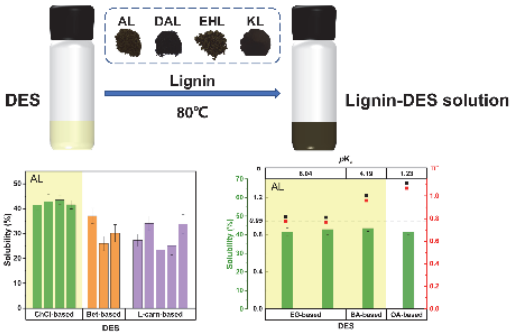
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Abstract: Lignocellulose utilization has the unique feature of net-zero carbon emissions. Thus, the utilization of lignocellulose as a renewable alternative to fossil-based carbon resources is one of the most promising strategies to achieve “carbon neutrality”. Lignin has been recognized as the most abundant aromatic biopolymer on Earth; hence, it could be the most promising alternative to fossil-based aromatics. The efficient dissolution of lignin is crucial for lignin upgrading, which relies on the design of innovative and robust solvents. Herein, we designed several biomass-derived acidic deep eutectic solvents (DESs) using choline chloride, betaine, and L-carnitine as hydrogen bond acceptors (HBAs), and four protic compounds as hydrogen bond donors (HBDs), namely, oxalic acid, benzoic acid, ethyl gallate, and 5-methoxysalicylic acid. The designed DESs can dissolve different types of lignin, including alkali lignin (AL), dealkaline lignin (DAL), enzymatic hydrolysis lignin (EHL), and Kraft lignin (KL). Lignin dissolution was found to be affected by the relative contents of three phenylpropanoid monomers in lignin: syringyl (S), guaiacyl (G), and *p*-hydroxyphenyl (H) units. More S and fewer H units in lignin could result in higher solubility. G-, S-, and H-type structural units were found in EHL, while AL, KL, and DAL had only G-type structural units. Therefore, EHL could be more easily dissolved than AL, KL, and DAL in the most developed DESs. The hydroxyl group content of the four lignin samples had a significant impact on lignin dissolution. AL (1.98 mmol·g^{-1}) and EHL (1.93 mmol·g^{-1}) had much higher contents of phenolic hydroxyl groups than DAL (0.62 mmol·g^{-1}), implying that AL and EHL had higher polarity than DAL. This resulted in different dissolution behaviors in different DESs with varying polarities. However, the sulfonate groups afforded KL with much higher polarity, thus resulting in the special dissolution behavior of KL. It is to be noted that not all cases of dissolving lignin in the developed DESs conformed to the above rules. Therefore, it is necessary to further explore the effect of the properties of the DESs on the dissolution of different lignins. Choline chloride was the preferred HBA to construct DESs with good performance and adaptability to lignin dissolution, whereas suitable acidity enabled benzoic acid and ethyl gallate to be favorable HBDs. Systematic investigation revealed that an efficient DES for lignin dissolution should possess stronger hydrogen-bonding acidity (α > 0.95) and appropriate polarity matching with the dissolved lignin. In addition, the pK_a value of the HBD and the acidity of the DESs were also efficient indices for estimating the performance of an acidic DES in dissolving lignin, and the pK_a value and acidity could be well correlated with the polarity. Generally, HBDs (e.g., BA and EG in this study) with moderate pK_a values can be employed to construct robust DESs to dissolve lignin with satisfactory solubility. Additionally, the viscosity of the DESs should have an



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impact on lignin dissolution, and a lower viscosity is helpful for dissolving lignin. Therefore, the better performance of the developed ChCl-based DESs than the Betaine- and L-Carnitine-based DESs was partially because of the lower viscosity of the ChCl-based DESs.

Key Words: Lignin dissolution; Deep eutectic solvents; Biomass-derived solvents; Hydrogen-bonding; Acidity; Polarity

生物质基酸性低共熔溶剂用于高效溶解木质素及溶解机理

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摘要：设计能高效溶解木质素的溶剂对木质素的高值化利用具有重要意义。本文中设计了基于氯化胆碱、甜菜碱和左旋肉碱作为氢键受体(HBA)和四种氢键供体(HBD)的生物质衍生的酸性低共熔溶剂(DESs)，可以溶解包括碱木质素(AL)、脱碱木质素(DAL)、酶解木质素(EHL)和硫酸盐木质素(KL)在内的不同类型的木质素。在大多数所设计的DESs中，EHL比AL、KL和DAL更容易溶解，而不同木质素中羟基的含量对木质素的溶解有显著影响，但是并非在所有DESs中木质素的溶解情况都符合上述规则。氯化胆碱是构建DESs的首选HBA，具有良好的性能并适应于不同类型木质素的溶解，而合适的酸度使苯甲酸和没食子酸乙酯成为对木质素溶解有利的HBDs。研究表明能有效溶解木质素的DESs应具有强的氢键酸度(α值 > 0.95)以及与溶解的木质素匹配的合适极性。此外，HBD的pK_a值和DESs的酸度也是评价酸性DESs溶解木质素性能的有效指标。通常具有适中pK_a值的HBDs能够用于构建具有高效的木质素溶解性能的DESs。DESs的粘度对木质素溶解也有一定影响，较低的粘度有助于木质素溶解。

关键词：木质素溶解；低共熔溶剂；生物质衍生溶剂；氢键；酸度；极性

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1 Introduction

Utilization of lignocellulose as a renewable alternative of fossil-based carbon resources (*i.e.*, coal, oil, and natural gas) is one of the most promising strategies to achieve “carbon neutrality” considering that lignocellulose utilization has the unique feature of net-zero carbon emission [1,2]. In this context, utilization of renewable and abundant-available biomass has gained worldwide interest [3,4]. As one of the three main components of lignocellulose, lignin has been recognized as the most abundant aromatic biopolymer on earth, thereby affording lignin being the most promising alternative to fossil-based aromatics (*i.e.*, chemicals and materials) [5–7]. Especially, the concept of “lignin-first biorefinery” significantly promotes the development of lignin upgrading to produce useful chemicals and functional materials [8–10]. Generally, dissolution of lignin in an appropriate solvent is unavoidable to realize the goal of effectively utilizing lignin. However, the complex cross-linked three-dimensional structure greatly hinders lignin dissolution [11]. To solve this challenging issue relays on designing robust and innovative solvent systems [12,13].

To date, various solvent systems have been developed to dissolve lignin, including aqueous organic solvents [14–17], liquid ammonia [18,19], ionic liquid (IL)-based systems [20–22], and deep eutectic solvents (DESs) [23–25]. Of these developed solvents, DESs are an emerging class of functional solvents, and are generally composed by two components, calling as hydrogen-bond donor (HBD) and hydrogen-bond acceptor (HBA), respectively [26–28]. In comparison with other solvent systems for lignin dissolution, DESs have some unique features. For example, DESs have lower volatility than organic solvents. Moreover, the process to prepare DESs is much simpler than that of synthesizing ILs, and organic solvents can be completely avoided for constructing DESs [27,29]. More importantly, the components in DESs can accept (for HBAs) or donate (for HBDs) hydrogen bonding, endowing DESs the ability to form hydrogen bonding with lignin, which is indispensable for dissolving lignin in a solvent [11,30]. These features make DESs be a highly promising class of solvents to dissolve lignin, and some DESs with good ability to dissolve lignin have been successfully developed. Generally, the DESs using protic chemicals (*i.e.*,

alcohols, carboxylic acids, amides, and phenols) as the HBDs showed stronger capability on dissolution of lignin [31–33]. However, the properties of lignin varied with the resources of raw lignocellulose and the extraction strategies to obtain lignin, thereby resulting in the unsatisfactory adaptability of the developed DESs on different types of lignin. Additionally, most of the developed DESs often involved fossil-based chemicals. Utilization of biomass-based derivatives to prepare DESs is a highly attractive strategy to achieve a "closed-loop" biorefinery. Recent works revealed that DESs could be directly prepared using biomass-derived chemicals, affording these DESs being biocompatibility and biodegradability. This feature of biomass-derived DESs enabled one-pot biomass processing with good environmental friendliness and cost effectiveness [34–37]. Although biomass-derived DESs have been prepared using choline chloride as the HBA and lactic acid or some lignin monomers as the HBDs for lignin dissolution or biomass pretreatment, the DESs prepared completely from renewable resources are still limited, and their capability of dissolving lignin needs to be further improved [23,38,39]. In spite of the encouraging achievements of using DESs to dissolve lignin, innovative and robust DESs completely derived from renewable resources are still highly desirable with the goal of making the dissolution processes green to the greatest extent. In another aspect, how to select HBAs and HBDs to construct robust DESs with high solubility of lignin is also an urgent issue to be solved. To date, there still lack reasonable methods or suitable standard to accurately evaluate the capability of one DES on lignin dissolution, resulting in that much time was wasted to select efficient DESs to dissolve lignin. Therefore, exploring a general rule for evaluation of DES's dissolution capability remains a great challenge.

With the purpose of developing robust renewable DESs to dissolve lignin and establish a reasonable index to evaluate the ability of acidic DESs to dissolve lignin, herein, we constructed several renewable acidic DESs using biomass-based HBAs and HBDs. In these designed DESs, choline chloride (ChCl), betaine (Bet), and L-carnitine (L-Carn) were selected as the renewable HBAs, while four biomass-based protic compounds (*i.e.*, oxalic acid (OA), benzoic acid (BA), ethyl gallate (EG), and 5-methoxysalicylic acid (5-MeOSA)) were employed as the HBDs. Four different types of lignin, including alkali lignin (AL), dealkaline lignin (DAL), Kraft lignin (KL), and enzymatic hydrolysis lignin (EHL), were selected to investigate the dissolution ability of the developed DESs. In these four types of lignin, AL was mainly derived from alkaline pulping black liquor and had a complex lignin condensation structure linked by C—C and ether bonds, while DAL was achieved from the residue obtained by alkali treatment of lignocellulose and subsequent acidizing separation. KL was usually produced by the reaction between lignin and lye when lignocellulose was cooked by the kraft method. The hydroxyl ions, sulfur ions and sulfhydryl ions in the kraft cooking liquor would effectively promote de-methoxide of lignin to obtain KL with more phenolic hydroxyl groups. EHL was extracted from the enzymatic residue of lignocellulose, which preserved the original structures (phenolic hydroxyl groups, and ether bonds) of the lignin. It was observed that the obtained DESs could dissolve various types of lignin. Notably, all the DESs using ChCl as the HBA possessed good performance on dissolving lignin, while BA and EG were preferred HBDs to generate robust DESs owing to their suitable acidity (based on the pK_a and the δ_P of trimethylphosphine oxide in ^{31}P NMR spectra). These results represented the major finding of this work, which was beneficial for designing robust renewable DESs to dissolve lignin.

2 Experimental section

2.1 Materials

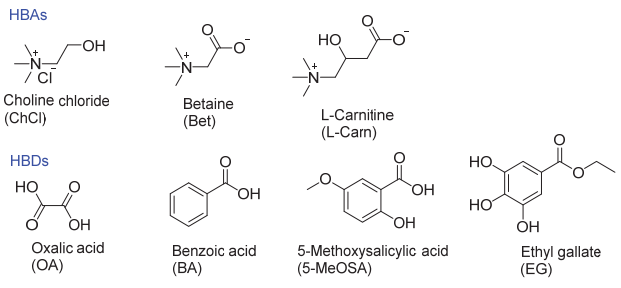
AL was extracted from softwood pulping black liquor, supplied by Shandong Longlive Bio-Technology Co., Ltd. (Shandong, China). The samples of EHL, DAL and KL were the same as those in our previous work [40]. Dealkaline lignin (DAL) was purchased from TCI Development Co., Ltd. (Shanghai, China). Kraft lignin (KL), 4-Nitroaniline (NH_2, 98%), and Nile red (NR, 97%) were obtained from Sigma-Aldrich Chemical Co., ltd. (Shanghai, China). Choline chloride (ChCl, 99%), L-carnitine (L-Carn, 98%), oxalic acid (OA, 99%), and trimethylphosphine oxide (TMPO, 97%) were purchased from Shanghai Aladdin Biochemical Co., Ltd. (Shanghai, China). Benzoic acid (BA, 99.5%), ethyl gallate (EG, 98%), and 5-methoxysalicylic acid (5-MeOSA, 97%) were purchased from Energy Chemical (Beijing, China). Betaine (Bet, 98%) was purchased from Beijing Innochem Technology Co., Ltd. (Beijing, China).

2.2 Preparation of DESs

The desired DESs were prepared by heating the mixture of suitable HBA and HBD (Scheme 1) with the molar ratio of 2 : 1, 1 : 1 or 2 : 1 at 100 °C in a glass vial until a homogeneous and transparent liquid was formed. All the obtained DESs were dried at 60 °C under vacuum for 24 h to remove the contaminated trace water, and then stored in a dryer for standby.

2.3 Determination of lignin solubility in DESs

The route to determine the solubility of lignin was similar with our previous work [23]. The experimental details were



Scheme 1 Chemical structures of the HBAs and HBDs to construct the target biomass-based DESs.

described as follows.

Initially, a desired DESs (*ca.* 1.0 g) was added into a glass flask of 10 mL and placed in an oil bath at a desired temperature. When the temperature was constant, a lignin sample of *ca.* 30 mg was added into the flask. After the lignin was completely dissolved, the above step was repeated until the added lignin could not be dissolved with a time of 2 h. Based on the amounts of the dissolved lignin and the DESs, a solubility with a higher potential deviation could be calculated.

Subsequently, according to the above results, lignin with a 90% amount of the above-achieved solubility was weighed and directly added into a glass flask of 10 mL containing the desired DESs of *ca.* 1.0 g at a desired temperature. After the added lignin was completely dissolved, the lignin was continually added with an increment size of *ca.* 5.0 mg until the lignin could not be dissolved within 2 h. Finally, the solubility with a lower potential deviation could be calculated.

2.4 Characterizations of lignin and DESs

2.4.1 Characterization of the desired biomass-derived DESs

Differential scanning calorimetry (DSC) was performed in a DSC Q2000 (TA Instruments Inc., USA) to determine the melting point of the obtained DESs. In the process of DSC examination, the sample was sealed in aluminum hermetic crucibles, and the temperature was raised from −80 °C to 100 °C with a heating rate of 10 °C·min^{-1} under nitrogen atmosphere. FT-IR spectra of these acidic DESs were recorded on a Tensor II spectrometer (Bruker, Germany) in the frequency range of 400–4000 cm^{-1}. The viscosity and conductivity of the acidic DESs were measured by Brookfield viscometer DV-II+ Pro and conductivity instrument (DDS-307A, INESA), respectively [41]. ^{31}P NMR spectra of the mixture of DES and TMPO were recorded on a Bruker Avance 600 MHz (Bruker, Germany) to evaluate the acidity of the developed DESs. The ^{31}P NMR samples were initially prepared by mixing the corresponding DES and TMPO with a molar ratio of 10 : 1, and then transferred into NMR tubes. To avoid the influence of deuterated reagents on the DES's properties, D_2O as an external lock was loaded in an inserted capillary tube. 85% H_3PO_4 aq. was used as the external reference at 0 ppm [42].

2.4.2 Characterization of lignin

All nuclear magnetic resonance (NMR) spectra of lignin were recorded on a Bruker Avance 400 MHz (Bruker, Germany) spectrometer. For 2D HSQC NMR spectra, about 50 mg of lignin was dissolved in 0.5 mL of DMSO-d_6, and the spectra were conducted according to previous literature [40]. For ^{31}P NMR spectra, 20 mg lignin sample was added into a solution of 500 μL anhydrous pyridine and deuterated chloroform (1.6 : 1, *v*/*v*). 100 μL Cyclohexanol (10.85 mg·mL^{-1} in anhydrous pyridine and deuterated chloroform, 1.6 : 1, *v*/*v*) was used as the internal standard and 100 μL of chromium (III) acetylacetonate solution (5 mg·mL^{-1} in anhydrous pyridine and deuterated chloroform 1.6 : 1, *v*/*v*) was employed as the relaxation reagent. Then, the mixture was reacted with 100 μL of phosphitylating reagent (2-chloro-4,4,5,5-tetramethyl-1,3,2-dioxaphospholate, TMDP) for about 10 min. ^{31}P NMR spectra were conducted according to our previous work [40].

3 Results and discussion

3.1 Preparation and characterization of DESs

The desired DESs in this work were constructed by heating the mixture of the HBAs and HBDs with a suitable molar ratio, and the detail route was described in the experimental section. DSC experiments indicated that the melting points of all the prepared DESs were below 100 °C (Table S1, and Figs. 1a and S1), which were much lower than the corresponding pure components. For instance, the melting points of L-Carn:5-MeOSA(1:1) and L-Carn:BA(1:1) were −16.85 °C and −27.55 °C, respectively, while the melting points of all the corresponding parent components were about 120 °C (Fig. 1a), confirming the formation of DESs from the selected HBAs and HBDs. The intermolecular hydrogen-bond interactions between HBAs and HBDs were investigated by FT-IR spectra (Figs. 1b and S2) [43]. It can be noted that the absorption peaks of O—H stretching vibration at 3300 cm^{-1} in EG changed to broader bands in the spectra of corresponding DESs, indicating the formation of highly associated hydrogen bonds in the prepared DESs. Besides, the thermal stability of acidic DESs were determined by TGA experiments [44]. The initial decomposition temperatures of almost all the prepared DESs were about 120 °C (Fig. 1c), which suggested that the DESs were stable under the conditions of dissolving lignin. In addition, the viscosity and

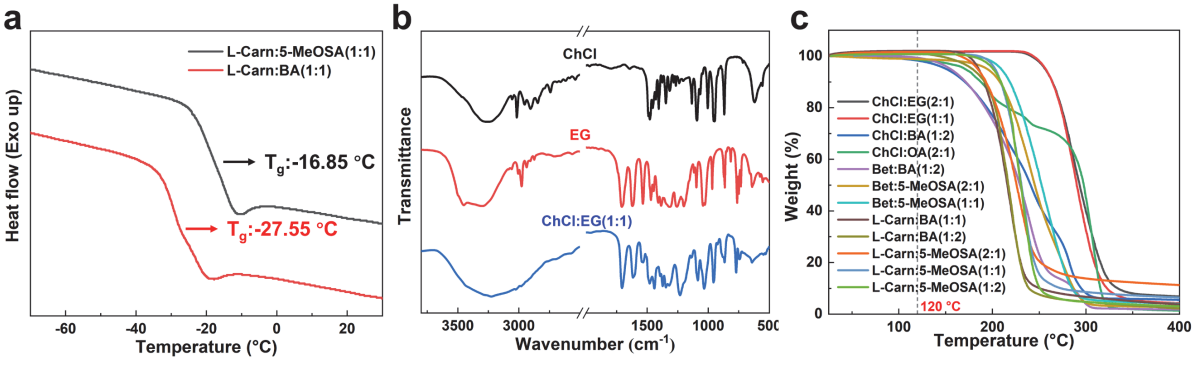


Fig. 1 DSC experiment of L-Carn-based DESs (a), FT-IR spectra of ChCl, EG, and ChCl-EG(1:1) (b), and TGA of the prepared DESs (c).

conductivity of these acidic DESs were also determined for their applications in lignin dissolution (Table S2).

3.2 Dissolution of AL in the prepared DESs

The ability of the prepared DESs to dissolve lignin was initially evaluated using the dissolution of AL as the model at 80 °C (Table 1). As shown in Table 1, all the synthesized ChCl-based DESs showed good ability to dissolve AL (Table 1, entries 1–4), and the difference on AL solubility (41.5%–43.5%, mass fraction hereafter) was negligible when the type of HBD or the ChCl/HBD molar ratio changed. When the HBA was changed into Bet (Table 1, entries 5–7), the highest solubility of AL was only 26.0% using Bet:5-MeOSA(2:1) as the solvent (Table 1, entry 6). In comparison, the ability of L-Carn-based DESs (Table 1, entries 8–12) to dissolve AL was better than that of Bet-based DESs, and the highest AL solubility in L-Carn-based DESs could reach up to 36.3% using L-Carn:5-MeOSA(1:2) as the solvent (Table 1, entry 12). Additionally, the effect of HBAs on AL solubility could be intuitively observed when DESs were generated using the three HBAs with BA at a molar ratio of 1 : 2 (Table 1, entries 3, 5, and 9), and the AL solubility decreased as follows: ChCl:BA(1:2) (43.5%) > Bet:BA(1:2) (37.1%) > L-Carn:BA(1:2) (34.1%). These results confirmed the advantage of ChCl to form DESs with the good ability to dissolve AL. Notably, for the DESs prepared using L-Carn and ChCl as the HBA, the AL solubility increased with the decreased HBA/HBD molar ratio, in other words, with the increase of HBD proportion, because more HBD had more sites to generate hydrogen bonding with AL, thereby affording the corresponding DESs having good ability to dissolve AL. For example, it was obvious that the AL solubility increased with the decrease of the L-Carn/5-MeOSA molar ratio as follows: L-Carn:5-MeOSA(2:1) < L-Carn:5-MeOSA(1:1) < L-Carn:5-MeOSA(1:2) (Table 1, entries 10–12). However, the relatively high melting points of Bet-based DESs resulted in the solubility in Bet:5-MeOSA(2:1) and Bet:5-MeOSA(2:1) did not follow the above rule (Table 1, entries 6 and 7), and the solubility was not much different.

3.3 Dissolution of other lignin (DAL, EHL, and KL) in the prepared DESs

Delighted by their excellent performance on AL dissolution, the prepared acidic DESs were applied to dissolve other three types of lignin, *i.e.*, DAL, EHL, and KL (Table S3) to investigate the adaptability of the developed DESs on different types of lignin. Similar with the dissolution of AL, all the ChCl-based DESs still showed good ability to dissolve DAL and EHL. Especially, in these ChCl-based DESs, ChCl:EG(1:1) showed the highest ability to dissolve DAL with a solubility of

Table 1 Dissolution of AL in the synthesized DESs [a].

Entry	DES	Molar ratio	AL solubility (%) [b]
1	ChCl:EG	2:1	41.5 (±2.2)
2	ChCl:EG	1:1	42.9 (±2.8)
3	ChCl:BA	1:2	43.5 (±1.7)
4	ChCl:OA	2:1	41.6 (±1.7)
5	Bet:BA	1:2	37.1 (±3.1)
6	Bet:5-MeOSA	2:1	26.0 (±2.9)
7	Bet:5-MeOSA	1:1	20.6 (±3.1)
8	L-Carn:BA	1:1	27.3 (±2.5)
9	L-Carn:BA	1:2	34.1 (±2.9)
10	L-Carn:5-MeOSA	2:1	23.5 (±2.7)
11	L-Carn:5-MeOSA	1:1	25.1 (±3.7)
12	L-Carn:5-MeOSA	1:2	33.9 (±3.8)

[a] The amount of DES was 1.0 g, and the dissolution temperature was 80 °C. [b] Values of solubility were the average of twice measurements, and the values in the parentheses were the standard deviations (%).

Table 2 Satisfactory results on dissolution of DAL, EHL, and KL in the corresponding DESs [a].

Entry	DES	lignin	Solubility (%) [b]
1	ChCl:EG(1:1)	DAL	49.4 (±2.5)
2	ChCl:BA(1:2)	EHL	44.4 (±1.5)
3	L-Carn:BA(1:1)	DAL	41.2 (±2.9)
4	L-Carn:BA(1:2)	DAL	44.4 (±2.7)
5	Bet:BA(1:2)	EHL	41.8 (±2.9)
6	L-Carn:BA(1:2)	EHL	41.6 (±2.9)
7	ChCl:EG(2:1)	KL	44.1 (±2.8)
8	ChCl:OA(2:1)	KL	62.0 (±1.9)
9	L-Carn:BA(1:2)	KL	39.2 (±1.8)

[a] The amount of DES was 1.0 g, and the dissolution temperature was 80 °C. [b] Values of solubility were the average of twice measurements, and the values in the parentheses were the standard deviations (%).

49.4% (Table 2, entry 1), while the highest solubility of EHL (44.4%) was obtained in ChCl:BA(1:2) (Table 2, entry 2). In comparison, lower solubility of DAL and EHL was achieved in most of the constructed Bet- and L-Carn-based DESs, and the exceptions were those using BA as the HBD. The solubility of DAL in L-Carn:BA(1:1) (Table 2, entry 3) and L-Carn:BA(1:2) (Table 2, entry 4) could reach 41.2 and 44.4%, respectively. Meanwhile, Bet:BA(1:2) (Table 2, entry 5) and L-Carn:BA(1:2) (Table 2, entry 6) showed high ability to dissolve EHL, and the EHL solubility was 41.8% and 41.6%, respectively. In addition, the dissolution behavior of KL in these developed DESs was obviously different from those of AL, DAL and EHL. Except for three DESs, *i.e.*, ChCl:EG(2:1) (Table 2, entry 7), ChCl:OA(2:1) (Table 2, entry 8), and L-Carn:BA(1:2) (Table 2, entry 9), other examined DESs showed unsatisfactory performance on KL dissolution. Surprisingly, a highest solubility of KL (62.0%) was obtained using ChCl:OA(2:1) as the solvent (Table 2, entry 8).

Furthermore, we examined the influence of dissolution temperature (60, 80, and 100 °C) on lignin solubility in seven optimized DESs with relatively good ability to dissolve all the four types of lignin (Fig. 2 and Table S4). Obviously, the solubilities of lignin increased with the rising of dissolution temperature. For all the determined DESs, the solubilities of the four types of lignin at 100 °C was higher than those at 80 °C. As well-accepted, the viscosity of DESs decreased with the increase of temperature, which was beneficial for the molecular interaction between the DESs and the lignin, thereby promoting the dissolution of lignin at higher temperature. In contrast, when the dissolution temperature decreased from 80 to 60 °C, the solubility of lignin significantly deceased because the greatly increased viscosity of DESs at lower temperature (e.g., 60 °C) was not conducive to the dissolution of lignin. In particular, the melting point of ChCl:BA(1:2) was near 60 °C, and Bet:BA(1:2) and Bet:5-MeOSA(1:1) became solid at 60 °C (Table S1), making them have no ability to dissolve lignin at this temperature. More specifically, ChCl:OA(2:1) could not dissolve DAL at 60 °C although the other three types of lignin (*i.e.*, AL, EHL, and KL) could be efficiently dissolved by ChCl:OA(2:1) at this temperature. This phenomenon may be related with the property of both DAL and ChCl:OA(2:1), which needed additional works to be revealed.

3.4 Effect of lignin structures on lignin dissolution in the developed DESs

As shown in Tables S3 and S4, the solubility in the developed DESs varied with the types of lignin, which was probably caused by their different structures. As well-accepted, there were two aspects in the molecular structures of different lignin samples to affect lignin dissolution. In terms of the resources, AL was extracted from softwood pulping black liquor, while EHL, DAL, and KL were the same with those in our previous report [40]. Thus, the structure of lignin was characterized by 2D HSQC NMR and ^{31}P NMR spectra, and the ^{31}P NMR results of EHL, DAL, and KL were achieved from the previous report [40]. First, the lignin dissolution was affected by the relative contents of three phenylpropanoid monomers in lignin, *i.e.*, syringyl (S), guaiacyl (G) and *p*-hydroxyphenyl (H) units. As reported, more S and fewer H units in lignin could result in higher solubility [45]. Based on the 2D HSQC NMR analysis of four types lignin (Fig. S3 and

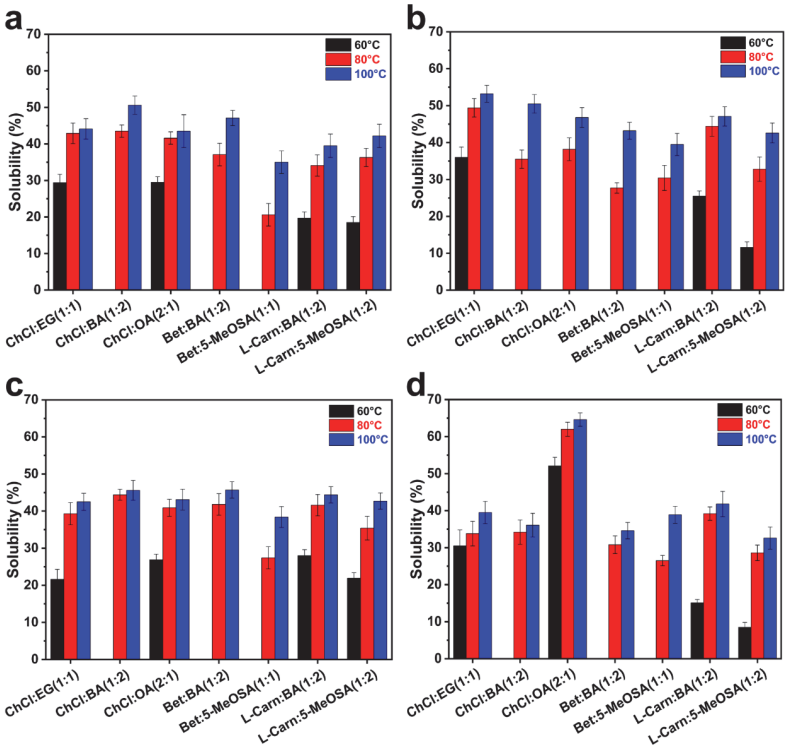


Fig. 2 Solubilities of different lignin in the prepared biomass-based acidic DESs. (a) AL, (b) DAL, (c) EHL, and (d) KL. The dark, red, and blue bars represent the experiments conducted at 60, 80, and 100 °C, respectively. Color online.

Table S5), G, S and H-type structural units were found in the EHL, while AL, KL and DAL had only G-type structural unit. Thereby, EHL could be more easily dissolved than AL, KL and DAL in most of the developed DESs. Second, the content of the hydroxyl groups in the four lignin samples had significant impact on lignin dissolution. Based on our previous report [40] and the quantitative ^{31}P NMR analysis of AL (Fig. S4), the contents of various hydroxyl groups in lignin were calculated (Table 3). As descried in Table 3, AL (1.98 mmol·g^{-1}) and EHL (1.93 mmol·g^{-1}) had much higher contents of phenolic hydroxyl groups than DAL (0.62 mmol·g^{-1}), implying that AL and EHL had higher polarity than DAL, thereby resulting in different dissolution behavior in different DESs with different polarity, which would be discussed in detail as follows. Because KL could not be completely dissolved in the phosphating agent, its ^{31}P NMR spectra was not determined. However, the contained sulfonate groups afforded KL having much higher polarity, thus making the special dissolution behavior of KL. Herein, it should be pointed out that not all the cases of dissolving lignin in the developed DESs conformed to the above rules. Therefore, it was necessary to further explore the effect of DESs' properties on the dissolution of different lignin.

3.5 Investigation of reasons on different performances of the developed DESs

From the experimental results in Tables 1 and 2, it could be deduced that ChCl was a preferred HBA to prepare efficient DESs for lignin dissolution considering high solubilities of various types of lignin (*i.e*., AL, DAL, EHL, and KL) in the developed ChCl-based DESs. Meanwhile, the DESs with BA or EG as the HBD generally showed good adaptability on various types of lignin. Originating from the synergistic effect of the examined HBAs and HBDs, ChCl:EG(2:1) provided satisfactory solubilities (> 40%) for all the four types of lignin (Table S3). As well-known, to break the intramolecular hydrogen bonding is essential to make lignin dissolved. Probably, the different performance of the developed DESs on lignin dissolution originated from their different ability to break intramolecular hydrogen bonding in lignin, which correlated with the properties of DESs, *i.e*., hydrogen-bonding acidity, hydrogen-bonding basicity, polarity, and dissociation degree (pK_a) or acidity.

Generally, Kamlet-Taft solvatochromic parameters (α, β, and π^*) are employed to evaluate the hydrogen-bonding acidity (α), hydrogen-bonding basicity (β), and polarity (π^*) of a solvent. Thus, we determined the Kamlet-Taft solvatochromic parameters of the prepared DESs (Table S6). After correlating with the solubility of lignin, it was observed that the DESs with higher α values (> 0.95) generally showed higher solubilities of lignin (Fig. 3). For example, ChCl:BA(1:2) and ChCl:OA(2:1) had much higher α values (1.2198, and 1.3534, respectively), and they showed better performance on dissolving all the determined lignin. As well-known, the higher the α values of the solvents are, the stronger ability of donating hydrogen bonding they have. Thereby, the DESs with higher α values had stronger ability to provide hydrogen bonding to lignin, which was beneficial for breaking the intramolecular hydrogen bonding in lignin, thus enhancing the dissolution of lignin. In contrast with the results of α values, no definite correlation was observed between the β value and the lignin solubility. For instance, the solubilities of the four types of lignin were all unsatisfactory in Bet:5-MeOSA(1:1) although it had the highest β value (2.0263). In comparison, the β value of ChCl:OA(2:1) was only −0.1677, but it showed good performance for all the types of lignin (Table S3). These results implied that the β value (hydrogen-bonding basicity) may be not a suitable index to evaluate the ability of DESs to dissolve lignin. Furthermore, we investigated the impact of polarity (described using π^* values) on lignin dissolution. When the DESs having larger π^* value, the DESs (e.g., ChCl:BA(1:2) and ChCl:OA(2:1)) could efficiently dissolve AL, EHL and KL, indicating that higher polarity was beneficial for dissolving these three types of lignin. Especially, higher polarity could more promote the dissolution of KL. Additionally, the solubilities of DAL in ChCl:BA(1:2) and ChCl:OA(2:1) were lower than those in ChCl:EG(2:1) and ChCl:EG (1:1) with lower π^* values. These results could be well explained by the principle of "Like Dissolves Like". Because the abundant sulfonic acid groups in KL significantly increased its polarity, ChCl:OA(2:1) with highest polarity provided the highest solubility of KL (62.0%). In comparison, as described above, the polarity of DAL was lower owing to the less hydroxyl and carboxylic groups in DAL (Table 3), thereby resulting in the relatively higher solubilities of DAL in ChCl:EG(2:1) and ChCl:EG(1:1) than those in ChCl:BA(1:2) and ChCl:OA(2:1) with larger polarity. Based on the results above, we could deduce that both stronger hydrogen-bonding acidity and matched polarity with the dissolved lignin could promote the dissolution of lignin in the developed DESs. The exception was the L-Carn:5-MeOSA(1:2), for which the solubility of lignin was lower than ChCl:BA(1:2) and ChCl:OA(2:1) despite of its higher α value (1.2138) and π^* value (0.9552), presumably due to the steric hindrance from the large molecular size of both L-

Table 3 Quantification of the functional groups of AL, DAL and EHL by quantitative ^{31}P NMR spectra.

Sample	Carboxylic group (mmol·g^{-1})	Aliphatic OH (mmol·g^{-1})	Phenolic OH (mmol·g^{-1})			Ref.
			Syringyl OH	Guaiacyl OH	p-Hydroxyphenyl OH	
AL	0.36	1.60	ND[a]	1.98	ND[a]	This work
DAL	ND[a]	1.15	ND[a]	0.62	ND[a]	40
EHL	0.97	1.18	0.61	0.74	0.58	40

[a] ND: Not Detected.

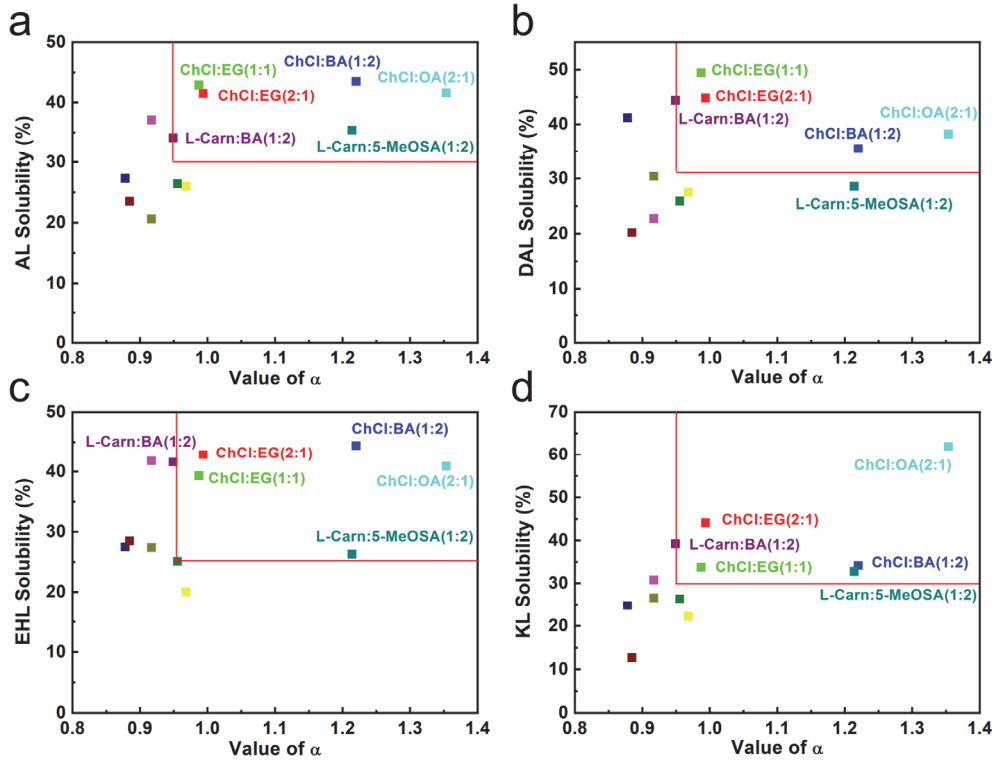


Fig. 3 Relationship between the value of *α* and the solubility of lignin.

(a) AL, (b) DAL, (c) EHL, (d) KL.

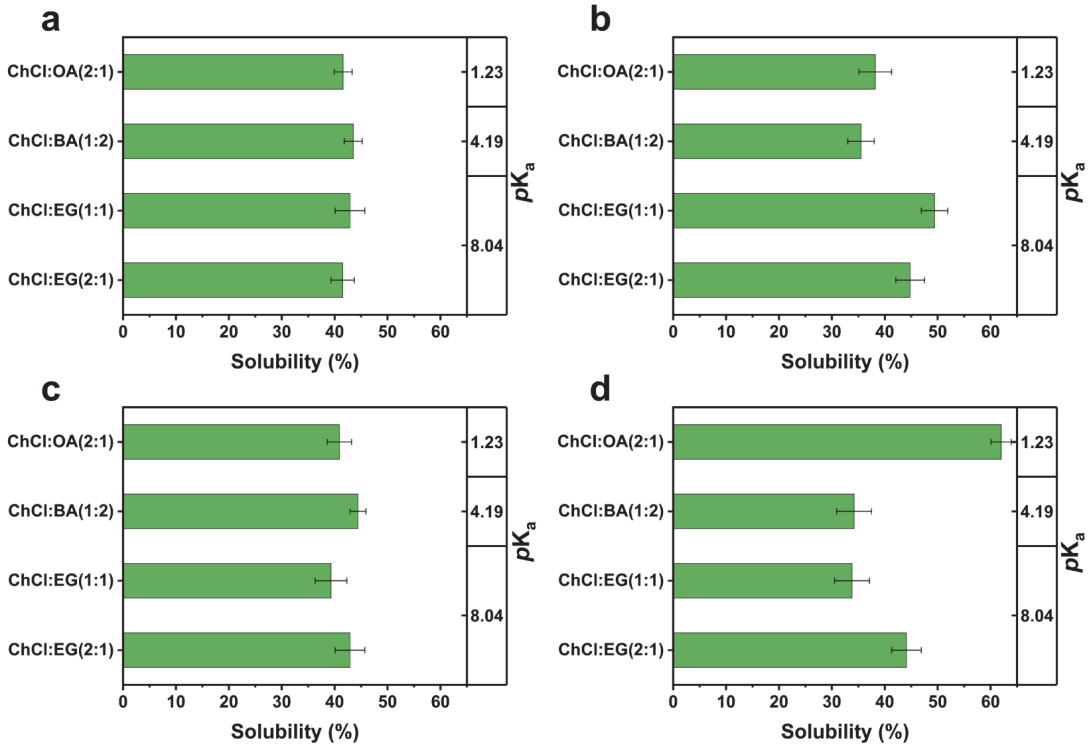


Fig. 4 Effect of pK_a of the used HBD on the solubility of lignin in ChCl-based DESs.

(a) AL, (b) DAL, (c) EHL, (d) KL.

Carn and 5-MeOSA as well as the high viscosity of L-Carn:5-MeOSA(1:2) (Table S2).

Considering the examined HBDs (*i.e.*, OA, BA, EG, and 5-MeOSA) were protic chemicals, we speculated that their dissociation degree, evaluated by the pK_a value, could also affect the performance of DESs on lignin dissolution. For ChCl-based DESs (Fig. 4 and Table S7), the DES (e.g., ChCl:OA(2:1)) using HBD with a low pK_a value showed good performance on KL

dissolution, while the DES (e.g., ChCl:EG) using HBD with a high pK_a value provided higher solubility of DAL. In comparison, the DESs (e.g., ChCl:BA) using HBD with a moderate pK_a value showed stronger ability to dissolve AL and EHL. For the Bet- and L-Carn-based DESs (Table S8), the difference on lignin solubility was not significant in most cases owing to the close pK_a value of the used HBDs (BA, or 5-MeOSA). These results were probably caused by that lower pK_a value indicated the higher dissociation degree of the HBDs, resulting in higher polarity of the corresponding DES. Based on the principle of "Like Dissolves Like", the DES using a HBD with lower pK_a value generally showed higher ability to dissolve lignin (e.g., KL) with more and strong polar groups (*i.e.*, hydroxyl, carboxylic, and sulfonic acid groups), while DAL with less polar groups could be well dissolved in the DES using a HBD with higher pK_a value. In comparison, the DESs using HBDs with the moderate pK_a value showed satisfactory ability to dissolve AL and EHL. Besides pK_a, the acidity of the DESs could be evaluated by the ^{31}P NMR chemical shift (δ_P) of trimethylphosphine oxide (TMPO) dissolved in the corresponding DES [46,47], and the DES resulting in larger δ_P value has stronger acidity. Thus, the δ_P values of several DESs were determined by ^{31}P NMR, and the results were provided in Fig. 5. After correlated with the solubility of lignin, it was observed that the difference on the lignin solubility was not significant when the δ_P values of the DESs were similar. Notably, the DES (ChCl:OA(2:1)) with the highest δ_P value showed the best performance on dissolution of KL, while the DES (L-Carn:BA(1:1)) with the lowest δ_P value showed bad ability to dissolve AL, EHL and KL. Nevertheless, L-Carn:BA(1:1) provided a satisfactory solubility of DAL. Generally, the δ_P values showed a positive correlation with the polarity. Thus, the influence of DES's acidity on lignin solubility showed similar tendency with the effect of DES's polarity, as discussed above.

From the discussions above, we could conclude that to achieve good solubility of lignin, the DES should simultaneously have the features of higher hydrogen-bonding acidity (the α value of > 0.95) and appropriate polarity (π^* value) to match the

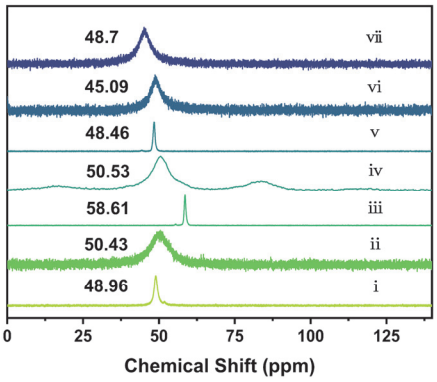


Fig. 5 ^{31}P NMR spectra of TMPO in DESs. (i) ChCl:EG(2:1), (ii) ChCl:EG(1:1), (iii) ChCl:OA(2:1), (iv) ChCl:BA(1:2), (v) Bet:BA(1:2), (vi) L-Carn:BA(1:1), and (vii) L-Carn:BA(1:2).

property of lignin. Besides, the pK_a value of the HBD and the acidity of the DESs were also efficient index to estimate the performance of an acidic DES to dissolve lignin, and the pK_a value and the acidity could be well correlated with the polarity. Generally, the HBDs (e.g., BA and EG in this work) with moderate pK_a value can be employed to construct robust DESs to dissolve lignin with the satisfactory solubility. Additionally, the viscosity of the DESs should have some impact on lignin dissolution, and lower viscosity was helpful for dissolving lignin. Thereby, the better performance of the developed ChCl-based DESs than Bet- and L-Carn-based DESs, partially resulting from the lower viscosity of ChCl-based DESs (Table S2).

3.6 Characterization of the regenerated lignin from the DES solution

The dissolved lignin could be easily regenerated from the biomass-derived acidic DESs (using ChCl:EG(1:1) as an example) with water as the anti-solvent, and the recovered lignin was named RAL, RDAL, RKL and REHL, respectively. The regenerated lignin was characterized by 2D HSQC NMR technique to examine the structure change after dissolution and regeneration process, and the results were described in Fig. S3. It was observed that the signal patterns of the native lignin and the corresponding regenerated one were rather similar, suggesting that the basic structure of the lignin did not change after the dissolution and regeneration process. Moreover, the relative abundances (X/100Ar) of different linkages were quantitatively determined by integrating the related signals (Fig. S3). Notably, all the regenerated lignin showed a decrease in the relative abundances of β-O-4 linkages in comparison with the native lignin, implying the cleavage of β-O-4 linkages during the dissolution and regeneration process. For example, the relative abundances of β-O-4 linkages in RKL were declined to 10.6/100Ar from 21.6/100Ar in the native KL. However, the decrease on the relative abundance of C—C bonds was much fewer compared with the decrease on the relative abundance of β-O-4 linkages, suggesting that the ability of the used acidic DES to cleave the C—C bonds was much lower than that of cleaving β-O-4 linkages. It could be deduced that the breakage of β-O-4 linkages was the main reaction during lignin dissolution in acidic DESs, which was a general phenomenon when lignin was dissolved in acidic DESs [11,39].

These results above implied that the developed acidic DESs could be potentially utilized to promote conversion of lignin into value-added chemicals. Besides, in previous reports, acidic DESs had been applied in lignin conversion. For example, Lopes *et al.* reported that dissolution of lignin in acidic DES favored the formation of lignin oligomers, which could be further convert into syringol, acetosyringone, vanillic and syringic acids [48]. Yuan *et al.* described that phenolic diketones could be obtained from lignin oil after acidic DES dissolution [49]. From the current experiments and the previously reported results, the developed acidic DESs could be highly promising for catalytic conversion

of lignin, which were undergoing in our lab. We did not investigate this aspect because lignin conversion was beyond the scope of the current work.

4 Conclusions

In conclusion, several renewable DESs were designed and constructed using biomass-derived chemicals as the HBAs (*i.e.*, ChCl, Bet, and L-Carn) and protonic HBDs (*i.e.*, OA, BA, EG, and 5-MeOSA). These prepared DESs could dissolve different types of lignin, including AL, DAL, EHL, and KL, and the solubility varied with the types of lignin and the DESs. Interestingly, the DESs prepared using ChCl as the HBA showed better performance and adaptability for different types of lignin. More importantly, the ability of the DESs to dissolve lignin correlated well with their hydrogen-bonding acidity and polarity. When a DES having stronger hydrogen-bonding acidity (α value of > 0.95) and appropriate polarity matching with the dissolved lignin, it would possess relatively good performance on lignin dissolution. Besides, it was observed that HBDs (e.g., BA and EG) or DESs with moderate acidity would be preferred to design robust DESs to dissolve lignin. We believe that the developed renewable acidic DESs have great potential in lignin dissolution and lignocellulose pretreatment. Notably, the finding that DESs with a strong hydrogen-bonding acidity, appropriate polarity, and moderate acidity generally have good ability to dissolve lignin is highly helpful to design innovative biomass-based DESs applied in lignocellulose utilization.

Supporting Information: available free of charge *via* the internet at http://www.whxb.pku.edu.cn.

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